

CHEW THE FAT

DESCRIPTION: Teams will demonstrate knowledge of the human digestive system and proper nutrition.

COMPETITORS BRING:

- Pencil/writing utensil;
- 3-ring binder of any size with any research/reference material. The binder may contain any item, such as a book, computer generated printout, or student created paperwork. No electronic materials are allowed. All materials must be securely bound inside the binder, so that when it is opened vertically (upside down) and given a light shake test no materials will fall out.

NUMBER OF PARTICIPANTS: 2

IMPOUND: NO

THE COMPETITION:

This event will be run in a station format. Teams will rotate through stations that assess any or all of the following topics:

1. Identify the major organs and body parts involved in the digestive process and understand the important job each body part has:
 - a. salivary glands
 - b. taste buds
 - c. teeth: incisors, premolars, molars, and canines
 - d. esophagus, stomach, liver, small intestine, large intestine
 - e. kidney, urine, bladder
2. Compare and contrast chemical and physical digestion.
3. Understand food and the benefits of vitamins & minerals.
 - a. Vitamin List: A, B, C, D, E, K
 - b. Minerals List: Calcium, Iron, Magnesium, Phosphorous, Sodium
4. Be able to test for the presence of different nutritional components in foods. For example:
 - a. Fats and oils using the paper bag test and to discover which foods contain high amounts of fats and oils.
 - b. Iron using a magnet in ground up breakfast cereals
 - c. Starch in foods using the iodine test
5. Know how to use food labels to make better food choices, specifically identifying serving size, vitamin and mineral content, overall caloric content plus identifying where the calories are coming from (fat, protein, carbohydrate).
6. Know common foods in the major food groups: fruits, vegetables, grains, protein, dairy and where they fit in the food pyramid and in the USDA My Food Plate.
7. Understand the relationship among the amount of food energy (calories) consumed, weight, and metabolism.

RELATED FLORIDA STANDARDS:

- SC.4.N.1.4 Attempt reasonable answers to scientific questions and cite evidence in support.
- SC.4.N.1.6 Keep records that describe observations made, carefully distinguishing actual observations from ideas and inferences about the observations.
- SC.5.N.1.1 Define a problem, use appropriate reference materials to support scientific understanding, plan and carry out scientific investigations of various types.
- SC.5.L.14.1 Identify organs in the human body and describe their functions, including the skin, brain, heart, lungs, liver, intestines, pancreas, muscles and skeleton, reproductive organs, kidneys, bladder, and sensory organs.

SCORING: High score wins. Points will be awarded for the accuracy of responses. Ties will be broken by the accuracy or quality of responses to pre-selected questions chosen by the event supervisor.

RESOURCES:

[MyPlate.gov | U.S. Department of Agriculture](#)

<https://www.nutrition.gov/topics/nutrition-life-stage/children/kids-corner>